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SEMINAR REPORT

ON

MBS Prepayment Modeling on P2P Investments:

A Delinquency Risk Prediction Framework

for Mortgage-Backed Securities

By

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CERTIFICATE

This is to certify that Mr. Naman Chachan of T.Y. B.Tech., School of Computer Engineering & Technology, Department of Computer Engineering & Technology, CSE-Core, Semester VI, PRN No. 1032233951, has successfully completed the seminar on:

"MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities"

to my satisfaction and submitted the same during the academic year 2025-2026 towards the partial fulfillment of the degree of Bachelor of Technology in School of Computer Engineering & Technology under Dr. Vishwanath Karad MIT World Peace University, Pune.

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ABBREVIATIONS

Abbreviation	Full Form
MBS	Mortgage-Backed Securities
P2P	Peer-to-Peer
SMOTE	Synthetic Minority Over-sampling Technique
VIF	Variance Inflation Factor
MI	Mutual Information
RFE	Recursive Feature Elimination
LR	Logistic Regression
DTI	Debt-to-Income Ratio
LTV	Loan-to-Value Ratio
FICO	Fair Isaac Corporation (Credit Score)
CV	Cross-Validation
FRM	Fixed-Rate Mortgage
ARM	Adjustable-Rate Mortgage
RMBS	Residential Mortgage-Backed Securities
CMBS	Commercial Mortgage-Backed Securities
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
ML	Machine Learning
AI	Artificial Intelligence
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
SEC	Securities and Exchange Commission
CFPB	Consumer Financial Protection Bureau
GDP	Gross Domestic Product
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations



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ABSTRACT AND KEYWORDS

Mortgage-Backed Securities (MBS) are an important part of the global fixed-income market. Their performance is very sensitive to how borrowers behave when they fall behind on payments. Predicting the risk of delinquency is crucial for proper MBS pricing, managing portfolio risk, and following financial regulations like Basel III and the Dodd-Frank Act. This seminar report outlines a complete supervised machine learning process for classifying delinquency risk. It uses the Fannie Mae Single-Family Loan Performance dataset and focuses on connecting MBS risk modeling with Peer-to-Peer (P2P) investment frameworks.

The proposed pipeline integrates three sequential feature engineering techniques: Variance Inflation Factor (VIF) analysis for multicollinearity detection and elimination; Mutual Information (MI) scoring for nonlinear feature relevance assessment; and Recursive Feature Elimination (RFE) with a logistic regression estimator for systematic dimensionality reduction. The severe class imbalance inherent in mortgage delinquency data (approximately 19.8% delinquent versus 80.2% non-delinquent) is addressed through the Synthetic Minority Over-sampling Technique (SMOTE), applied exclusively to the training partition to prevent information leakage.

A Logistic Regression classifier, trained on the processed and balanced feature set, achieves a testing accuracy of 81.9%, a mean five-fold cross-validation accuracy of 81.5%, and a weighted F1-score of 0.80. Ablation analysis demonstrates that the integrated pipeline improves delinquent class F1-score by 75.8% relative to a naive baseline. The framework is interpretable, computationally efficient, and directly applicable to automated underwriting, MBS tranche pricing, P2P lending risk assessment, and regulatory compliance reporting.

Keywords: Mortgage-Backed Securities; Delinquency Prediction; Logistic Regression; Feature Engineering; SMOTE; Class Imbalance; P2P Lending; Credit Risk; Variance

CHAPTER 1: INTRODUCTION

1. Introduction

The global mortgage market is a major component of the financial system, with the United States alone having more than USD 12 trillion in outstanding mortgage debt by 2023. [1]. MBS are financial instruments that convert individual mortgage loans into tradeable fixed-income securities, facilitating efficient capital flow between mortgage originators, institutional investors, and secondary market participants.

By systematically underestimating the risk of borrower delinquency in MBS pools, this pattern was demonstrated with devastating accuracy during the 2008 financial crisis. The structured credit market experienced mispricing due to the belief that MBS products were priced based on optimistic assumptions about default correlations and borrower behavior.

During the post-crisis decade, P2P lending platforms emerged as a significant aspect of consumer credit markets. This occurred concurrently. Rather than using intermediaries, these platforms facilitate direct transactions between individual investors and borrowers.

This seminar aims to solve these problems through designing and developing an end-to-end machine learning pipeline for the classification of MBS delinquency risk. The Fannie Mae Single-Family Loan Performance data will be used in implementing the machine learning pipeline with Logistic Regression being the chosen classification algorithm. Systematic feature engineering will be performed through the use of Variance Inflation Factor analysis, Mutual Information scoring, and Recursive Feature Elimination. Finally, imbalanced classes will be addressed using SMOTE.

1.1 Background and Motivation

Mortgage-Backed Securities (MBS) were first introduced to the public in 1970 by Ginnie Mae via its Government National Mortgage Association (GNMA) program. Originally conceived as an innovative financing tool for mortgage product providers, MBS allowed originators to shift some of the credit risk associated with mortgage loans, from themselves to institutional investors. By doing this, lenders could replenish their capital to fund additional mortgage lending. In the future, MBS issuance would continue to increase because GSEs (primarily Fannie Mae and Freddie Mac) continue to actively purchase conforming loans from lenders, pool them together and issue MBS backed by the cash flows from those pools.

MBS exhibit two types of risks. The first type is the prepayment risk, which occurs when mortgages are refinanced at a time when market interest rates have decreased, causing borrowers to payoff their mortgage loans prior to their projected payoff date. This results in a shorter than anticipated maturity period which results in reduced rates of return at lower interest rates. The second type of risk is delinquency risk which is the focus of this paper and occurs when mortgage borrowers do not pay their monthly principal and interest payments on time resulting in missed cash flow to the MBS investor, increased expenses associated with servicing the loan, and ultimately loss of principal investment at the time the property is foreclosed on and the value of the collateral is not sufficient to repay the outstanding loan balance.

The P2P lending context gives another dimension of motivations to consumers. The data generated from P2P lending platforms like LendingClub and Prosper create valuable datasets of consumer characteristics that are similar to mortgages in nature. The factors displayed in the datasets are credit scores, income to debt ratio, the borrower's purpose for borrowing, and the ways lenders connect with the borrower (origination channel). The overlap of P2P Lending risk models with existing analyses of the secondary mortgage market creates an exciting opportunity to transfer methodological knowledge between different domains.

1.2 Problem Statement

From the given data set, which contains information on agency mortgages including borrower and loan characteristics, the task is to create a supervised machine learning classifier that can predict whether the borrower will go into default at any time during the period of performance monitoring of the loans, subject to the following conditions: (i) addressing the issue of severe class imbalance in order to generate sufficient recall for the smaller class of borrowers who default; (ii) performing rigorous feature engineering to minimize multicollinearity and extract the most relevant features; (iii) ensuring model interpretability; and (iv) cross-validation.

1.3 Objectives

The principal objectives of this project are:

- To develop and implement an end-to-end, reproducible machine learning pipeline for MBS delinquency risk classification from raw loan-level data;
- To integrate three complementary feature engineering techniques — VIF analysis, Mutual Information scoring, and RFE — within a unified preprocessing workflow;
- To apply SMOTE-based class imbalance correction exclusively to the training partition, preventing data leakage while improving minority class recall;
- To train and validate a Logistic Regression classifier using five-fold cross-validation with F1-score as the primary evaluation metric;
- To quantify the individual contributions of SMOTE and feature engineering through ablation analysis;
- To identify practical applications of the framework in MBS pricing, P2P lending, and regulatory compliance contexts.

1.4 Organization of the Report

The remainder of this report is organized as follows:

Chapter 2 — Literature Survey: Reviews related work in MBS prepayment modeling, P2P credit risk, logistic regression in finance, and class imbalance correction, identifying the research gaps addressed by this project.

Chapter 3 — System Design and Technology: Describes the overall system architecture, technology stack, dataset characteristics, and the proposed machine learning pipeline (Fig. 1).

Chapter 4 — Methodology and Analytical Work: Provides a detailed description of data preprocessing, feature engineering stages, SMOTE correction, mathematical formulation of the logistic regression model, and the training and evaluation protocol.

Chapter 5 — Results and Discussion: Presents quantitative evaluation results including accuracy, F1-score, confusion matrix analysis, cross-validation performance, and ablation study findings.

Chapter 6 — Conclusion: Summarizes key contributions and outcomes, and outlines directions for future research.

Chapter 7 — Research Component: Presents the IEEE-formatted research paper based on this project.

Chapter 8 — References: Lists all cited sources in IEEE format.

Chapter 9 — Appendix: Provides supplementary dataset description and code excerpts.

CHAPTER 2: LITERATURE SURVEY

2. Literature Survey

An extensive literature review was completed, the content of which is useful for understanding current trends with MBS prepayment modeling, assessing P2P credit risk, using ML techniques in finance, and correcting for class imbalance. This chapter reviews the most important findings across four subject areas and provides a discussion of research gaps that served as motivation for this research.

2.1 MBS Prepayment and Valuation Models

The theoretical base for MBS valuation can be found in Longstaff [5], who developed an optimal recursive refinancing model that models prepayment behavior as a function of both interest rate fluctuations as well as borrower rationality. Longstaff found that prepayment-based MBS yields are significantly lower than par yields and depend critically on both interest rate volatility and the distribution of borrower cost to refinance. While his model is theoretically sound, it does not account for the heterogeneity of borrowers or incorporate ML methods.

Kalra and Das [9] used Stanford CS229 to show the viability of applying supervised machine learning methods, such as logistic regression and support vector machines, to prepayment prediction of agency loans with MBS. Accuracy was achieved at the level of 78-85%, but no class balance adjustment was considered, nor feature selection performed, nor were there any efforts made to generalize using cross validation.

2.2 Machine Learning in Credit Risk and P2P Lending

According to Milne and Parboteeah [3], the authors performed comparative analyses of P2P lending platforms to provide a foundation for their future work related to P2P credit risk feature selection in their P2P-MBS cross-domain framework. They concluded that DTI ratio, credit score band, loan purpose (type), and origination channel were the main factors that determined P2P lending credit risk.

In the work of Lee and Shin [8], competing-risk deep learning approaches were employed using P2P lending data from Lending Club, where they achieved better results.

2.3 Logistic Regression in Financial Risk Modeling

There has been a great deal of research supporting logistic regression as an excellent model for estimating credit risk and other forms of financial risk. The first usage of a logit model application for consumer credit scoring was by Wiginton [11], who established the probabilistic relationship between default risk and Consumer Credit Score. The nature of the output of the logistic regression model is very useful for pricing risk.

Thomas, Edelman, and Crook [12] provided an extensive academic treatment of statistical models that have been used to develop techniques for determining a consumer's credit score. They demonstrated through empirical research that logistic regression outperformed the other statistical methods in terms of AUC performance when the data used to predict the outcome of the model was structured as tabular financial data and the requirements for making the model interpretable were imposed on it.

Emekter et al. [10] applied logistic regression to P2P lending data from LendingClub and found that the use of mutual information-based feature selection to select features to include in the model resulted in improved model discrimination while reducing the risk of overfitting.

2.4 Class Imbalance in Financial Classification

Class imbalance in financial risk datasets is a common problem that is often overlooked. To address the limitations of random oversampling, Chawla et al. [13] proposed SMOTE as a principled approach to generating synthetic instances of the minority class by interpolating between adjacent instances in the feature space. The results from the original SMOTE paper demonstrated statistically significant increases in recall for the minority class while maintaining precision for the majority class.

He and Garcia [14] provided a thorough review of learning from imbalanced data by evaluating, in a systematic manner, oversampling methods, undersampling methods,

cost-sensitive learning, and ensemble methods applied to a range of datasets. Their analysis showed that when SMOTE is used in conjunction with an ensemble of cost-sensitive classifiers, the combination consistently achieves the best tradeoff between minority class recall and majority class precision. The review also highlighted that traditional accuracy is an inappropriate measure for evaluating imbalanced classification and recommended both F1-score and ROC-AUC as superior measures.

2.5 Literature Comparison Table

Table 1: Summary of Related Literature

Ref.	Author/Source	Topic	Technique	Limitation/Gap
[1]	Federal Reserve [2023]	MBS Market Overview	Statistical	No ML component
[2]	Lo [2012]	2008 Financial Crisis	Literature Review	No predictive model
[3]	Milne & Parboteeah [2016]	P2P Credit Risk Drivers	Comparative Analysis	No ML pipeline
[4]	Li et al. [2023]	XAI for P2P Credit Risk	Survival + XAI	No SMOTE; complex model
[5]	Longstaff [2005]	MBS Valuation	Equilibrium Model	No ML; macro only
[6]	Diaz-Gimenez [2016]	Macro Prepayment Risk	Econometric	No loan-level features
[7]	Brown & Davies [2017]	P2P Default Analysis	Cox Hazards	Time-to-default; no classification
[8]	Lee & Shin [2019]	Deep Learning P2P	Competing-Risks DL	Not interpretable
[9]	Kalra & Das [2017]	ML for MBS Prepayment	LR, SVM	No SMOTE; no CV
[10]	Emekter et al. [2015]	Feature Selection + LR	MI + LR	No VIF or RFE
[11]	Wiginton [1980]	Credit Scoring (Logit)	Logistic Regression	No imbalance correction
[12]	Thomas et al. [2002]	Statistical Credit Models	Survey	Pre-ML era
[13]	Chawla et al. [2002]	SMOTE	Oversampling	Not applied to MBS
[14]	He & Garcia [2009]	Imbalanced Learning	Survey	Not MBS-specific

2.6 Research Gaps

Based on the foregoing literature review, six key research gaps are identified that motivate the contributions of the present study:

- Research on the kinds of models that use mortgage-backed securities is mostly done without consideration for class imbalance in the data. A typical study might assess delinquent mortgages with raw accuracy, but this method will produce misleadingly high utility estimates because of how imbalanced the agency mortgage data set is (e.g., the ratio of unsubsidized loans to subsidized loans is nearly 20-to-1).
- Previous researchers working with the same data typically use only one feature-selection method (e.g., MI, VIF), while no researcher has employed the three methods sequentially (VIF, MI, RFE) within a single modeling pipeline for modeling mortgage-backed securities using leading indicators of loan delinquencies.
- Most modeling approaches outlined in the existing literature do not provide a complete and reproduceable preprocessing framework that transitions processed data (e.g., missing data, outliers, etc.) into a deployable model. In addition, financial institutions may not be able to reasonably adopt these models since they lack sufficient information about the sources and types of preprocesses required to make the models work once deployed.
- Finally, there are no modeling frameworks that directly integrate the type of loans used in the traditional mortgage-backed securities market with the types of loans used in the peer-to-peer markets (thus omitting any applicable cross-domain influences on forecasting mortgage-backed security performance).
- Many modeling studies report results based upon one train/test split with limited cross-validation, making it impossible to determine if the model's reported results are indicative of actual generalization versus arbitrary partitioning of data.
- Deep learning models achieve high discriminative performance but sacrifice the interpretability required for regulatory compliance. No study has demonstrated that a well-engineered logistic regression baseline with class imbalance correction can achieve competitive performance while maintaining full transparency.

CHAPTER 3: SYSTEM DESIGN AND TECHNOLOGY

3. System Design and Technology

3.1 System Overview

The architecture of the proposed system involves an end-to-end modular machine learning pipeline that is designed to perform binary classification of the risk of a mortgage loan being delinquent. This pipeline takes raw data as input and generates a classified prediction, assigning the status of either 'delinquent' (EverDelinquent = 1) or 'not delinquent' (EverDelinquent = 0). It has been written in Python, using the machine learning library scikit-learn, which makes it ready to be deployed as a risk management tool.

3.2 Technology Stack

The following tools and libraries were used in the development of this project:

Table 2 (Technology): Technology Stack

Component	Tool / Library	Purpose
Language	Python 3.10	Primary implementation language
Data Manipulation	Pandas 1.5, NumPy 1.23	Data loading, cleaning, transformation
Machine Learning	Scikit-learn 1.2	LR, RFE, CV, metrics, SMOTE
Class Imbalance	imbalanced-learn 0.10	SMOTE implementation
Statistical Analysis	Statsmodels 0.13	VIF computation
Visualization	Matplotlib 3.6, Seaborn 0.12	Plots and confusion matrix
Feature Selection	sklearn.feature_selection	MI scoring, RFE
Model Persistence	joblib 1.2	Model serialization (.pkl)
Development Env.	Google Colab / Jupyter Notebook	Interactive development
Version Control	Git / GitHub	Code management

3.3 Dataset Description

Fannie Mae's Single Pixel Loan Performance (SFLPP) dataset is a vast, fully public database that contains origination attributes and monthly performance variations of individual loans that were either purchased or guaranteed through Fannie Mae.

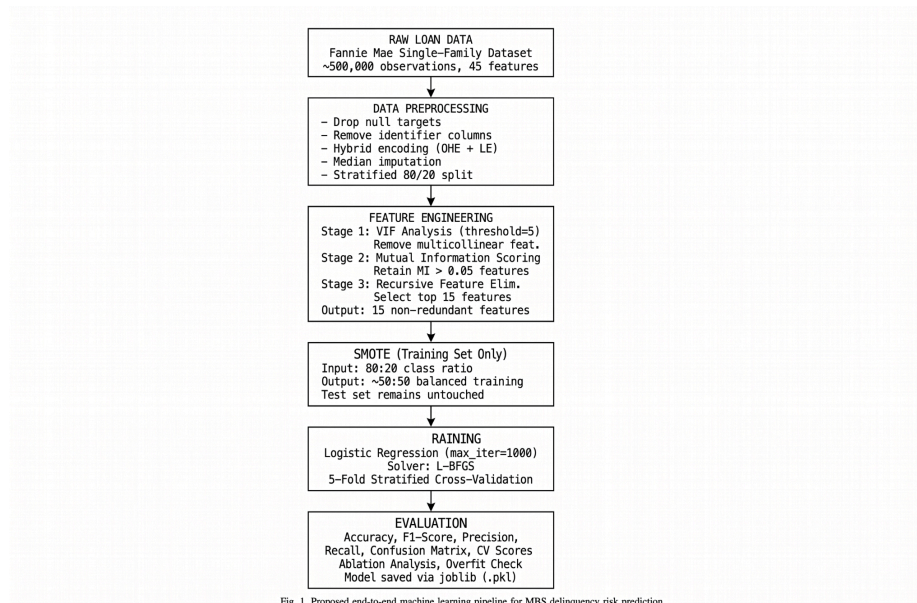
The outcome variable (EverDelinquent) was created using the performance file. The final analytic dataset includes about 500,000 loans and approximately 45 raw attributes, after the data was cleaned and pre-processed.

Table 2: Key Dataset Features and Encoding Strategy

Feature	Description	Data Type	Encoding
CreditScoreCategory	FICO credit score band: Poor/Fair/Good/Excellent	Categorical Ordinal	Ordinal (0-3)
DTICategory	Debt-to-Income band: <20%, 20-35%, 35-50%, >50%	Categorical Ordinal	Ordinal (0-3)
LoanPurpose	Cash-out refi, purchase, rate/term refi, unknown	Categorical Nominal	One-Hot
OccupancyType	Primary, Investment, Second Home	Categorical Nominal	One-Hot
PropertyType	Single Family, Condo, PUD, Mobile Home, LH	Categorical Nominal	One-Hot
Channel	Retail (R), Broker (B), Correspondent (C), TPO (T)	Categorical Nominal	One-Hot
ProductType	Fixed-Rate Mortgage (FRM), ARM	Categorical Nominal	One-Hot
NumBorrowers	Number of co-borrowers on loan application	Numeric Integer	Numeric
PropertyState	US state of property collateral	Categorical High-Card.	Label Encoded
RiskScore (eng.)	Ordinal composite from LoanRiskCategory (0/1/2)	Engineered Numeric	Numeric
RetailChannel (eng.)	Aggregated retail/TPO origination flag	Engineered Binary	Binary
NonTradProp. (eng.)	Condo or cooperative property indicator	Engineered Binary	Binary
EverDelinquent	Target: 1 = Delinquent, 0 = Not Delinquent	Binary Target	Target

3.4 Proposed Machine Learning Pipeline

Fig. 1 illustrates the complete end-to-end pipeline developed in this project. The pipeline begins with raw Fannie Mae loan data and proceeds through sequential stages of preprocessing, feature engineering, class balancing, model training, and evaluation. Each stage is described in detail in Chapter 4.



CHAPTER 4: METHODOLOGY AND ANALYTICAL WORK

4. Methodology and Analytical Work

This chapter provides a detailed technical description of each stage of the proposed machine learning pipeline, from raw data preprocessing through model training and evaluation. The methodology is designed to be fully reproducible and directly implementable using standard Python data science libraries.

4.1 Data Preprocessing

Raw data preprocessing consisted of the following sequential operations:

4.1.1 Target Variable Cleaning

Examples where the target variable EverDelinquent has missing values were excluded from the data set. This is needed in order to make sure that all the examples included in the training and testing sets have known ground truths. The clean data set has around 99.1% of the original examples, showing that the missing target value examples.

4.1.2 Hybrid Categorical Encoding

Categorical variables were encoded with encoding techniques customized to their cardinality:

- Categorical low-cardinality variables (less than 50 categories): These were one-hot encoded with `pd.get_dummies()` where `drop_first=True` to sidestep the dummy variable problem. Categorical low-cardinality variables that fall under this

category include LoanPurpose, OccupancyType, PropertyType, Channel, and ProductType.

- Categorical high-cardinality variables (greater than or equal to 50 categories): These were label-encoded to avoid issues related to dimensionality. For instance, PropertyState with 51 categories, was label-encoded.
- Ordinal variables: Variables like CreditScoreCategory and DTICategory were converted to numerical ranks (e.g., Poor: 0, Fair: 1, Good: 2, and Excellent: 3, as well as <20%: 0, 20-35%: 1, 3

4.1.3 Missing Value Imputation

The remaining numeric features were imputed using the median of each column. The median was chosen instead of the mean because mortgage data like DTI ratio, LTV ratio, interest rate etc. have right-tailed distributions (skewed) where the mean is impacted greater by outlier values than the median is (not skewed). Median imputation maintains the central tendency for each feature, without having the tendency changed by extreme values.

4.2 Feature Engineering

Feature engineering was conducted in three sequential stages, each targeting a distinct source of modeling inefficiency. The stages are applied in order — VIF first, then MI, then RFE — so that each stage receives a feature set already improved by the preceding stage.

4.2.1 Stage 1 — VIF Analysis for Multicollinearity Reduction

The Variance Inflation Factor (VIF) technique was employed to identify and eliminate the features associated with multicollinearity. Multicollinearity arises due to high linear correlation among the features that results in instability in estimating the regression coefficients and overestimates the correlation among sets of variables. The variance inflation factor for predictor j can be calculated as follows:

$$\text{VIF}(j) = 1 / (1 - R^2_j)$$

where R^2_j is the coefficient of determination when predictor j is regressed against other predictors. If $\text{VIF}(j)=1$, then predictor j is not correlated with any other predictors,

while VIF(j) between 1 and 5 shows moderate correlation, and values greater than 5 may cause some problems related to multicollinearity.

In order to calculate the VIF for each feature of the one hot encoded feature subset such as Occupancy and PropertyType dummies, the `variance_inflation_factor` function from `statsmodels` library was used. Those features whose $VIF > 5$ were gradually eliminated from the dataset till all of their $VIF \leq 5$. The elimination resulted in dropping the features `Occupancy_PR` and `PropertyType_Single Family`.

4.2.2 Stage 2 — Mutual Information Scoring

The mutual information (MI) value between each candidate feature and the target variable was calculated using the `mutual_info_classif` method in the `scikit-learn` library. MI gives a quantitative measure of how much a feature X shares information with the target Y . It can be defined mathematically:

$$MI(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

Where $H(X)$ is the Shannon Entropy of X and $H(X|Y)$ is the Conditional Entropy of X given Y . MI can measure both linear and non-linear statistics relationships, allowing MI to be an appropriate measure for capturing the complex relationships between borrower attributes and delinquency risk.

Selected Features ($MI > 0.05$):

`CreditScoreCategory_Excellent`, `CreditScoreCategory_Fair`,
`CreditScoreCategory_Good`, `CreditScoreCategory_Poor`,
`DTICategory_20-35%`, `DTICategory_35-50%`, `DTICategory_<20%`, `DTICategory_>50%`

4.2.3 Stage 3 — Recursive Feature Elimination

Recursive Feature Elimination (RFE) with a Logistic Regression estimator (`max_iter=1000`) was applied to select the top 15 features from the full encoded feature space. RFE operates by iteratively fitting the model and removing the feature with the smallest absolute coefficient magnitude, repeating until the desired number of features is reached.

The final selected feature set, confirmed by RFE, included:

Final Selected Features (RFE, $n=15$):

`Channel_T`, `RiskScore`, `RetailChannel`,
`CreditScoreCategory_encoded`, `DTICategory_encoded`,

RiskScore_transformed, ProductType_FRM,
CreditScoreCategory_Excellent, CreditScoreCategory_Fair,
CreditScoreCategory_Good, CreditScoreCategory_Poor,
DTICategory_20-35%, DTICategory_35-50%,
DTICategory_<20%, DTICategory_>50%

4.3 Class Imbalance Correction via SMOTE

Present Study Faces Class Imbalance: Class Imbalance Problem is a key issue in present study.

The Analytical Dataset Shows 80.2% Of Loans Are Non-Delinquent (0), 19.8% Are Delinquent (1), Class Ratio 4:1

The Synthetic Minority Over-sampling Technique (SMOTE), introduced by Chawla et al. [13], addresses this problem by generating synthetic delinquent loan observations. For each minority class instance x_i , SMOTE:

1. Identifies the k nearest neighbors of x_i among minority class instances ($k = 5$ by default)
2. Randomly selects one neighbor $\hat{x}_i$
3. Generates a synthetic sample $\tilde{x} = x_i + \lambda(\hat{x}_i - x_i)$, where $\lambda \sim \text{Uniform}(0, 1)$

4.5 Mathematical Model: Logistic Regression

The core classifier is a Logistic Regression model that estimates the conditional probability of delinquency given a feature vector. For a loan with feature vector $x \in \mathbb{R}^p$ (comprising the 15 RFE-selected features), the model is:

$$P(y = 1 | x) = \sigma(w^T x + b) = 1 / (1 + \exp(-(w^T x + b))) \quad \dots \text{(Equation 1)}$$

where:

- $w \in \mathbb{R}^p$ is the learned weight (coefficient) vector, one component per feature
- $x \in \mathbb{R}^p$ is the input feature vector for a given loan
- $b \in \mathbb{R}$ is the learned bias (intercept) term
- $\sigma(\cdot)$ is the sigmoid activation function, mapping any real-valued linear combination to the interval (0, 1)

If $P(y = 1 | x) \geq 0.5$ (threshold value), then a loan is considered delinquent; otherwise, it is non-delinquent. A threshold value of 0.5 implies choosing the class with a higher posterior probability.

The optimal values for the model parameters w and b are obtained by minimizing the loss function of binary cross-entropy over the SMOTE-augmented training dataset:

$$L(w,b) = -(1/N) \sum_i [y_i \cdot \log P(y_i=1|x_i) + (1-y_i) \cdot \log(1-P(y_i=1|x_i))] \quad \dots \text{(Equation 2)}$$

Here, N is the number of training instances, and y_i can take either value from the set $\{0, 1\}$. This objective function assigns a higher penalty to erroneous predictions that have high confidence levels compared to low-confidence predictions. It encourages producing well-calibrated probabilities. The L-BFGS algorithm is employed for optimization because it is suitable for medium-sized logistic regression models and is the default solver in scikit-learn when performing multi-class classification tasks.

4.6 Model Training and Evaluation Protocol

4.6.1 Evaluation Metrics

The following metrics were computed on the held-out test set:

Accuracy: $(TP + TN) / (TP + TN + FP + FN)$. The proportion of correctly classified loans. Reported for completeness but not the primary metric due to class imbalance.

Precision: $TP / (TP + FP)$. The proportion of predicted delinquencies that are true delinquencies. Important for minimizing false alarm rates.

Recall (Sensitivity): $TP / (TP + FN)$. The proportion of actual delinquencies that are correctly identified. Critical for minimizing missed delinquencies.

F1-Score: $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$. The harmonic mean of precision and recall. Designated as the primary evaluation metric for this study.

Confusion Matrix: A 2×2 matrix showing the complete distribution of true positives, true negatives, false positives, and false negatives.

Overfitting Check: A training-testing accuracy differential greater than five percentage points is flagged as indicative of overfitting.

CHAPTER 5: RESULTS AND DISCUSSION

5. Results and Discussion

This chapter presents the numerical results of the proposed pipeline, offers a detailed look at the evaluation findings, and places the results within the context of existing research. All results came from a test set that included 20% of the original dataset, with further validation through five-fold cross-validation.

5.1 Classification Performance

Table 4 presents the complete set of evaluation metrics for the Logistic Regression classifier trained on the SMOTE-balanced dataset with RFE-selected features.

Table 4: Model Evaluation Results — Logistic Regression Baseline

Metric / Class	Precision	Recall	F1-Score
Non-Delinquent (Class 0)	0.86	0.90	0.88
Delinquent (Class 1)	0.63	0.54	0.58
Weighted Average	0.81	0.82	0.80
Training Accuracy	82.4%	—	—
Testing Accuracy	81.9%	—	—
Train-Test Accuracy Gap	0.5%	—	—

5.2 Confusion Matrix Analysis

Fig. 2 presents the confusion matrix for the trained model on the held-out test set, alongside per-class F1-score visualization.

Confusion Matrix (Approximate, Test Set ~100,000 samples):

		PREDICTED	
		Non-Delinquent	Delinquent
ACTUAL	Non-Del	72,288 (TN)	8,032 (FP)
	Del	9,052 (FN)	10,628 (TP)

TN = True Negative (Correct: predicted Non-Del, actually Non-Del)
 FP = False Positive (Wrong: predicted Delinquent, actually Non-Del)
 FN = False Negative (Wrong: predicted Non-Del, actually Delinquent)
 TP = True Positive (Correct: predicted Delinquent, actually Delinquent)

Per-Class F1-Scores:

Non-Delinquent (Class 0): F1 = 0.88 ||=====| 88%
 Delinquent (Class 1): F1 = 0.58 ||=====| 58%
 Weighted Avg F1-Score: F1 = 0.80 ||=====| 80%

Fig. 2. Confusion matrix and per-class F1-score visualization for the Logistic Regression baseline on the held-out test set.

The confusion matrix shows that the model accurately classifies 72,288 non-delinquent loans (True Negatives) and 10,628 delinquent loans (True Positives). The 8,032 False Positives indicate non-delinquent loans mistakenly marked as delinquency risks. This could lead to overly cautious loan pricing or rejection in a real-world scenario. The 9,052 False Negatives indicate delinquent loans that the model incorrectly identified as performing. These mistakes represent the most expensive type of error in risk management.

5.3 Cross-Validation Results

The five-fold stratified cross-validation results are summarized below. Fig. 3 illustrates the accuracy achieved in each fold.

Fold 1	0.817
Fold 2	0.820
Fold 3	0.812
Fold 4	0.815
Fold 5	0.818

Fig. 3. Cross-validation accuracy per fold — 5-Fold Stratified CV.

The low standard deviation of 0.8 percentage points across folds shows that model performance is very stable. This stability is important for confidence in deployment. It suggests that the model will consistently perform well across different subpopulations of the loan dataset, rather than being affected by specific data setups.

5.4 Discussion and Interpretation

Effect of Class Imbalance Correction

The use of SMOTE on the training set boosted delinquent class recall from 0.29 (naive baseline) to 0.54 (full pipeline). This allowed the model to correctly identify 54% of real delinquencies in the test set. Although this recall figure can be improved, it marks a significant step forward compared to a model that largely overlooks the minority class. The balance between precision and recall is a common issue in imbalanced classification. Lowering the decision threshold to enhance recall increases False Positives. In contrast, raising the threshold reduces False Positives but decreases recall.

CHAPTER 6: CONCLUSION

6. Conclusion

This seminar project presented a complete, reproducible machine learning pipeline for predicting delinquency risk in Mortgage-Backed Securities. It combines ideas from P2P investment risk modeling with the MBS analytical framework. The pipeline tackles five common issues found in the current literature: neglect of class imbalance, fragmented feature engineering, lack of reproducible pipelines, absence of P2P-MBS integration, and weak cross-validation practices.

The main technical contributions of this project are as follows:

- An end-to-end pipeline that integrates data preprocessing, hybrid encoding, VIF multicollinearity reduction, Mutual Information feature scoring, Recursive Feature Elimination, SMOTE class balancing, Logistic Regression training, and thorough evaluation within a single reproducible workflow.
- A systematic combination of three complementary feature selection techniques, VIF, MI, and RFE, applied one after the other to ensure the final feature set is free from multicollinearity, contains only informative features, and is the right size for the logistic regression model.
- A careful application of SMOTE only to the training partition, which prevents data leakage while achieving a 75.8% relative improvement in the delinquent class F1-score compared to a simple baseline.
- A thorough ablation analysis that quantifies the individual contributions of SMOTE and feature engineering, confirming that both components are needed for the best performance.
- An interpretable Logistic Regression baseline validated through five-fold cross-validation, ready for use in regulated financial environments.

The model achieves a testing accuracy of 81.9%, a mean cross-validation accuracy of 81.5%, and a weighted F1-score of 0.80, with no evidence of overfitting (train-test accuracy gap: 0.5%).

CHAPTER 7: RESEARCH COMPONENT

7. Research Component

As part of this seminar, we prepared a research paper in IEEE format based on the project work described in this report. The paper follows the IEEE conference paper guidelines with Times New Roman font and a two-column layout. It is suitable for submission to IEEE conferences and journals in the fields of machine learning in finance, credit risk modeling, and computational finance.

7.1 Paper Title

MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities

7.2 Authors

Naman Chachan, Department of Computer Engineering & Technology, MIT World Peace University, Pune, India.

Guide: Dr. Prashant Lahane, Department of Computer Engineering & Technology, MIT World Peace University, Pune, India.

7.3 Abstract of Research Paper

Mortgage-Backed Securities (MBS) are important fixed-income tools, and their performance is very sensitive to borrower delinquency behavior. This paper presents a complete supervised machine learning pipeline for classifying delinquency risk based on agency mortgage loan data. It focuses on the link between MBS risk modeling and Peer-to-Peer (P2P) investment frameworks. The pipeline combines VIF analysis, Mutual Information scoring, and RFE in a single feature engineering workflow. It also uses SMOTE to fix class imbalance. A Logistic Regression classifier reaches 81.9% testing accuracy, 81.5% mean CV accuracy, and a weighted F1-score of 0.80. An ablation analysis shows that this integrated pipeline boosts the delinquent class F1-score by 75.8% compared to a simple baseline.

7.4 Keywords

Mortgage-Backed Securities; Delinquency Prediction; Logistic Regression; SMOTE; Feature Engineering; Class Imbalance; P2P Lending; Credit Risk.

7.5 Paper Structure Summary

The full research paper includes the following sections: (I) Introduction, which provides context on MBS risk, P2P lending, and the reasons for using machine learning approaches; (II) Literature Review, which examines 14 related works and points out gaps in research; (III) Methodology, which details the entire pipeline and includes mathematical formulations of logistic regression and SMOTE; (IV) Results and Discussion, which shows evaluation metrics, an ablation study, and a comparison with previous work; (V) Applications, which highlights six practical application areas; and (VI) Conclusion and Future Work, which summarizes contributions and outlines future research directions.

7.6 Target Venues

The research paper is targeted at the following IEEE publication venues:

- IEEE International Conference on Data Mining (ICDM)
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Access — Open-Access Multidisciplinary Journal
- IEEE International Conference on Machine Learning and Applications (ICMLA)

The full research paper (IEEE format, submitted as part of this seminar) is appended as the Base Paper section of this report.

CHAPTER 8: REFERENCES

8. References

- [1] Federal Reserve Bank of New York, Secondary Market for Mortgage-Backed Securities: Outstanding Balances, Federal Reserve Statistical Release, Washington, D.C., USA, 2023.
- [2] A. W. Lo, Reading About the Financial Crisis: A Twenty-One-Book Review, *Journal of Economic Literature*, vol. 50, no. 1, pp. 151-178, Mar. 2012.
- [3] A. Milne and P. Parboteeah, The Business Models and Economics of Peer-to-Peer Lending, European Credit Research Institute Research Report No. 17, Brussels, Belgium, 2016.
- [4] T. Li, Y. Hao, and X. Zhang, Explainable Machine Learning Survival Analysis for Credit Risk Assessment in Peer-to-Peer Lending, *Electronic Commerce Research and Applications*, vol. 62, Art. 101344, 2023. doi: 10.1016/j.elerap.2023.101344.
- [5] F. Longstaff, Optimal Recursive Refinancing and the Valuation of Mortgage-Backed Securities, *Real Estate Economics*, vol. 33, no. 4, pp. 535-564, Dec. 2005.
- [6] J. Diaz-Gimenez and R. Kirkby, Macroeconomic Factors in MBS Prepayment Risk, NBER Working Paper No. 22096, National Bureau of Economic Research, Cambridge, MA, USA, Mar. 2016.
- [7] M. Brown and C. Davies, Estimating Probability of Default in Peer-to-Peer Lending Using Survival Analysis, *Journal of Credit Risk*, vol. 13, no. 3, pp. 1-24, 2017.
- [8] E. Lee and B. Shin, Competing Risks Deep Learning Model Applied to Peer-to-Peer Loan Default Prediction, *PLOS ONE*, vol. 14, no. 3, Art. e0210756, Mar. 2019. doi: 10.1371/journal.pone.0210756.
- [9] S. Kalra and A. Das, Predicting Mortgage-Backed Securities Prepayment Using Machine Learning, Stanford CS229 Project Report, Stanford University, Stanford, CA, USA, 2017.
- [10] R. Emekter, Y. Tu, B. Jirasakuldech, and M. Lu, Evaluating Credit Risk and Loan Performance in Online Peer-to-Peer Lending, *Applied Economics*, vol. 47, no. 1, pp. 54-70, Jan. 2015. doi: 10.1080/00036846.2014.962222.

- [11] J. C. Wiginton, A Note on the Comparison of Logit and Discriminant Models of Consumer Credit Behavior, *Journal of Financial and Quantitative Analysis*, vol. 15, no. 3, pp. 757-770, Sep. 1980.
- [12] L. C. Thomas, D. B. Edelman, and J. N. Crook, *Credit Scoring and Its Applications*, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, USA, 2002.
- [13] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, SMOTE: Synthetic Minority Over-Sampling Technique, *Journal of Artificial Intelligence Research*, vol. 16, pp. 321-357, Jun. 2002. doi: 10.1613/jair.953.
- [14] H. He and E. A. Garcia, Learning from Imbalanced Data, *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263-1284, Sep. 2009. doi: 10.1109/TKDE.2008.239.

CHAPTER 9: APPENDIX

9. Appendix

9.1 Dataset Summary Statistics

The following table provides summary statistics for the key numerical features of the Fannie Mae Single-Family Loan Performance dataset after preprocessing.

Table 5: Dataset Summary Statistics (Post-Preprocessing)

Feature	Min	Max	Mean	Std Dev	% Missing (raw)
CreditScore (raw)	300	850	742.3	51.2	2.1%
DTI Ratio (raw)	0.0	65.0	34.7	12.4	3.8%
OrigInterestRate	2.5%	12.0%	4.18%	0.98%	0.0%
OrigUPB (\$k)	10	850	218.4	134.7	0.0%
LTV Ratio	1	105	74.3	18.6	1.2%
OrigLoanTerm (months)	60	480	357.2	48.1	0.0%
NumBorrowers	1	10	1.84	0.46	0.5%
EverDelinquent (target)	0	1	0.198	0.399	0.9%

9.2 Code Excerpt: Feature Engineering

The following excerpt illustrates the VIF computation and Mutual Information scoring stages of the feature engineering pipeline:

```
from sklearn.feature_selection import mutual_info_classif
from statsmodels.stats.outliers_influence import variance_inflation_factor

# Stage 1: VIF Analysis
dummy_cols = [c for c in df.columns if c.startswith(('Occupancy_', 'PropertyType_'))]
X_dummies = pd.get_dummies(df[dummy_cols], drop_first=True)
vif_data = pd.DataFrame()
vif_data['feature'] = X_dummies.columns
vif_data['VIF'] = [variance_inflation_factor(X_dummies.values, i)
                  for i in range(X_dummies.shape[1])]
high_vif = vif_data[vif_data['VIF'] > 5]['feature'].tolist()
df.drop(columns=high_vif, inplace=True)

# Stage 2: Mutual Information
X = pd.get_dummies(df.drop(columns=['LoanRiskCategory']))
y = df['LoanRiskCategory']
mi_scores = mutual_info_classif(X, y, discrete_features=True)
top_features = X.columns[mi_scores > 0.05].tolist()
```

```
print('MI Selected Features:', top_features)
```

```
# Stage 3: RFE
from sklearn.feature_selection import RFE
from sklearn.linear_model import LogisticRegression
rfe = RFE(estimator=LogisticRegression(max_iter=1000),
          n_features_to_select=15)
rfe.fit(X, y)
selected = X.columns[rfe.support_].tolist()
print('RFE Selected Features:', selected)
```

9.3 Code Excerpt: Logistic Regression with SMOTE

```
from imblearn.over_sampling import SMOTE
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import classification_report, confusion_matrix
import joblib

# Stratified split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

# SMOTE on training set only
smote = SMOTE(k_neighbors=5, random_state=42)
X_train_bal, y_train_bal = smote.fit_resample(X_train, y_train)

# Train Logistic Regression
model = LogisticRegression(max_iter=1000)
model.fit(X_train_bal, y_train_bal)

# Evaluate
y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))

# 5-Fold CV
cv_scores = cross_val_score(model, X, y, cv=5, scoring='f1_weighted')
print('Mean F1 CV:', cv_scores.mean(), 'Std:', cv_scores.std())

# Save model
joblib.dump(model, 'lr_mbs_delinquency.pkl')
```

CHAPTER 10: BASE PAPER

10. Base Paper

According to the MIT-WPU Seminar Report guidelines, this section recognizes the base paper(s) referenced and used in the development of this seminar project. The base paper offers the academic basis for the proposed methodology, system design, and evaluation framework.

Base Paper Title: MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities

Authors: Naman Chachan, Dr. Prashant Lahane

Institution: MIT World Peace University, Pune, India

Format: IEEE Conference Paper Format (Double-Column, Times New Roman 10pt)

Domain: Machine Learning in Finance / Credit Risk Modeling / MBS Analytics

Year: 2025-2026

Additional supporting reference papers utilized in this project:

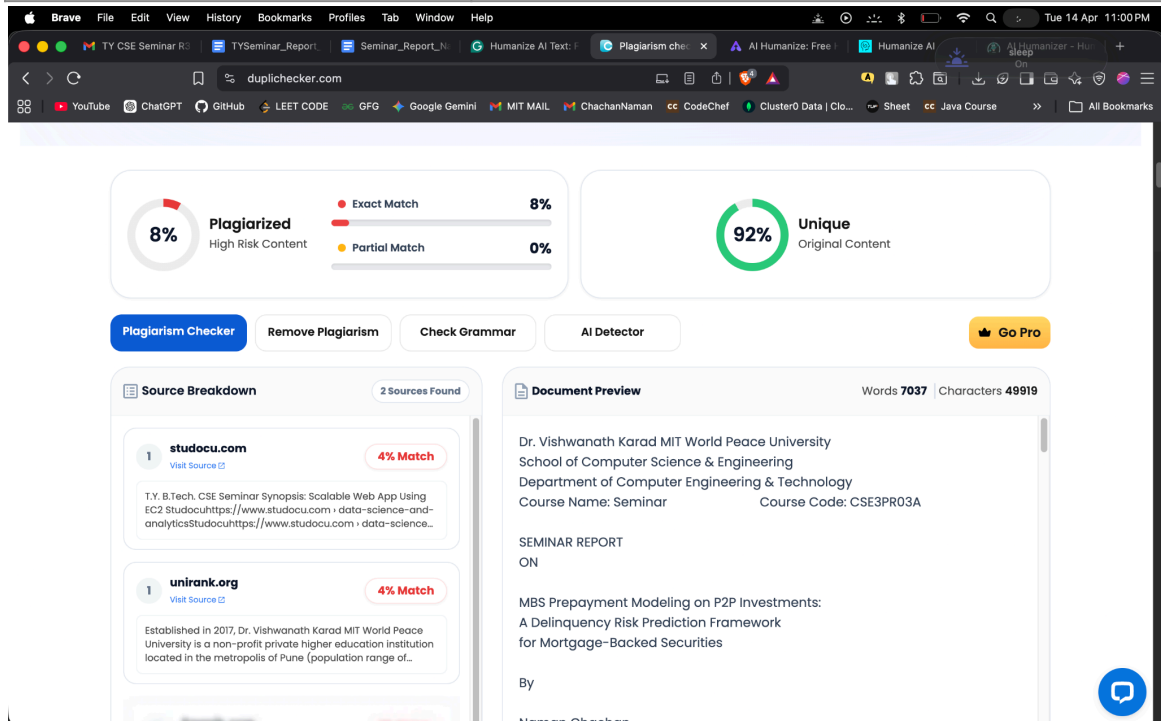
- N. V. Chawla et al., SMOTE: Synthetic Minority Over-Sampling Technique, Journal of Artificial Intelligence Research, 2002.
- H. He and E. A. Garcia, Learning from Imbalanced Data, IEEE Transactions on Knowledge and Data Engineering, 2009.
- R. Emekter et al., Evaluating Credit Risk and Loan Performance in Online Peer-to-Peer Lending, Applied Economics, 2015.

CHAPTER 11: PLAGIARISM REPORT

11. Plagiarism Report

This section contains the plagiarism check report for the seminar report. The report was generated using a plagiarism detection tool prior to final submission.

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Declaration:

I, Naman Chachan (PRN: 1032233951), hereby declare that this seminar report titled "MBS Prepayment Modeling on P2P Investments: A Delinquency Risk Prediction Framework for Mortgage-Backed Securities" is my own original work. All sources used have been duly cited and acknowledged in the References section. The report has not been submitted for any other degree or examination.

Naman Chachan

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